

Assignment of IPK Project 1 – Multiple variants (OMEGA/DELTA)

Deadlines

Date	Description
22. 3. 2025 23:59	Registrations end
23. 3. 2025 23:59	Submission deadline

Evaluation

You can get up to 7 points for working applications in compliance with the assignment, as well as demonstrating good programming skills.

You can get up to 3 points for documentation.

You can earn extra points for significant extension of your program's features.

You cannot get more than 10 points from the project, but the points for extra effort can help you minimize any losses during the assesment of standard requirements.

Languages

- C
- C++
 - Object-oriented programming **is mandatory**
- C#
 - Object-oriented programming **is mandatory**

Variants

- ALPHA: Last year's points from Project 2
 - only points from the last academic year 23/24 are accepted;
 - last year's score $x / 20 \times 10$;
- [DELTA: L2/L3 scanner](#)

- [OMEGA: L4 scanner](#)

OMEGA Variant: L4 Scanner

Assignment

1. Create a simple TCP and UDP network L4 scanner. The program will scan the specified hostname or IP address(es) (plural IP addresses in the case of multiple answers to DNS query) and ports. It will output to stdout port statuses (open, filtered, closed) (7 pts.)
2. Create relevant manual/documentation for the project (3 pts.)

Specification

The application scans the selected ports of device (translated onto one or more IPv4/IPv6 addresses) on a given network interface.

Packets/Frames should be sent using sockets. If needed, you can eavesdrop on the responses using the libpcap library.

The program can be terminated at any given moment with `Ctrl + C` sequence.

Scanning should be done and return results as fast as possible. During development and testing, try scanning only the computers you own or manage.

TCP scanning

Sends only SYN packets. It does not perform a complete 3-way-handshake. If an RST response is received - the port is marked as *closed*. If no response is received from the scanned port after a given time interval, it must be verified with another packet and only then the port is marked as *filtered*. If a service is running on the port, the port is marked as *open*. See RFC 793 for more information.

UDP scanning

With UDP scanning, you can think of a given computer responding with an ICMP message of type 3, code 3 (port unreachable) when the port is *closed*. Consider the other ports as *open*.

Execution

```
./ipk-l4-scan [-i interface | --interface interface] [--pu port-ranges |  
--pt port-ranges | -u port-ranges | -t port-ranges] {-w timeout} [hostname  
| ip-address]
```

```
./ipk-l4-scan --help

./ipk-l4-scan --interface

./ipk-l4-scan
```

where:

- `-h / --help` writes usage instructions to `stdout` and terminates
- `-i eth0` (just one interface to scan through) or `--interface` . If this parameter is not specified (and any other parameters as well), or if only `-i / --interface` is specified without a value (and any other parameters are unspecified), a list of active interfaces is printed (additional information beyond the interface list is welcome but not required).
- `-t` or `--pt` , `-u` or `--pu` port-ranges - scanned tcp/udp ports, allowed entry e.g., `--pt 22` or `--pu 1-65535` or `--pt 22,23,24` . The `--pu` and `--pt` arguments can be specified separately, i.e. they do not have to occur both at once if the user wants only TCP or only UDP scanning
- `-w 3000` or `--wait 3000` , is the timeout in milliseconds to wait for a response for a single port scan. This parameter is optional, in its absence the value 5000 (i.e., five seconds) is used.
- either `hostname` , or `ip-address` , which either is hostname (e.g., merlin.fit.vutbr.cz) or IPv4/IPv6 address of scanned device.
- All arguments can be in any order.

Execution Examples

```
./ipk-l4-scan --interface eth0 -u 53,67 2001:67c:1220:809::93e5:917
./ipk-l4-scan -i eth0 -w 1000 -t 80,443,8080 www.vutbr.cz
```

Functionality Illustration

```
./ipk-l4-scan -i eth0 --pt 21,22,143 --pu 53,67 localhost
```

```
127.0.0.1 21 tcp closed
127.0.0.1 22 tcp open
127.0.0.1 143 tcp filtered
127.0.0.1 53 udp closed
127.0.0.1 67 udp open
```

Output Format

The application is going to be subject to automated testing. It is of utmost importance for the application to write the result to `stdout` exactly as specified.

Program output (`stdout`) consists of one or more lines. Individual lines can be in any order. Each line contains values separated by space: scanned IP address (IPv4 or IPv6), port number, protocol type (`tcp` or `udp`), port state (`open` , `filtered` , or `closed`).

```
./ipk-l4-scan -i eth0 -t 22 localhost
```

```
127.0.0.1 22 tcp open
```

```
./ipk-l4-scan -i eth0 -t 21,22 -u 22,21 localhost
```

```
127.0.0.1 21 tcp closed
```

```
127.0.0.1 22 tcp open
```

```
127.0.0.1 21 udp closed
```

```
127.0.0.1 22 udp closed
```

```
./ipk-l4-scan -i eth0 -t 80 www.fit.vutbr.cz
```

```
147.229.9.23 80 tcp open
```

```
2001:67c:1220:809::93e5:917 80 tcp open
```

Bibliography

- RFC 793: Transmission Control Protocol, 1981. Online. Request for Comments. Internet Engineering Task Force. [Accessed 17 February 2025].
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DELTA Variant: L2/L3 Scanner

Assignment

1. Create a simple network ICMP(v6), ARP/NDP scanner in C/C++/C#. The program discovers what devices are available from a selected range of IP addresses. It prints to standard output the availability status of the given IP addresses at the L2 and L3 layers (7 pts.)
2. Create relevant manual/documentation for the project (3 pts.)

Specification

The application scans for presence of L2 and L3 devices on given network segment(s).

Packets/Frames should be sent using raw sockets. If needed, you can eavesdrop on the responses using the libpcap library.

The program can be terminated at any given moment with `Ctrl + C` sequence.

Scanning should be done and return results as fast as possible. During development and testing, try scanning only the computers you own or manage.

Execution

```
./ipk-l2l3-scan [-i interface | --interface interface] {-w timeout} [-s
ipv4-subnet | -s ipv6-subnet | --subnet ipv4-subnet | --subnet ipv6-
subnet]
```

```
./ipk-l2l3-scan --help
```

```
./ipk-l2l3-scan --interface
```

```
./ipk-l2l3-scan
```

where:

- `-h / --help` writes usage instructions to `stdout` and terminates
- `-i eth0` (just one interface to scan through) or `--interface` . If this parameter is not specified (and any other parameters as well), or if only `-i / --interface` is specified without a value (and any other parameters are unspecified), a list of active interfaces is printed (additional information beyond the interface list is welcome but not required).
- `-w 3000` or `--wait 3000` , is the timeout in milliseconds to wait for a response for a single port scan. This parameter is optional, in its absence the value 5000 (i.e., five seconds) is used.
- `-s 192.168.1.0/24` or `--subnet fd00:cafe:0000:face::0/120` specifies which segments to scan using IPv4 or IPv6. There can be multiple segments to be scanned (i.e., the `-s/--subnet` argument can be repeated when the program is called).
- The application must be able to infer the correct network address and the resulting number of hosts to be scanned from the user input of the `-s` or `--subnet` argument.
- The application does not have to deal with the "bloat" of the `-s` or `--subnet` argument input with respect to the number of hosts being scanned (e.g., too short netmask or prefix length, for instance `-s 10.0.0.0/8`) or the location of the segment being scanned (i.e., attempting to ARP scan a network to which the computer is not directly connected).
- All arguments can be in any order.

Execution Examples

```
./ipk-l2l3-scan -i eth0 -w 1000 -s 192.168.0.0/25 -s 192.168.128.0/29
./ipk-l2l3-scan --interface eth0 --wait 1000 --subnet fd00:cafe:0000:face::0/120
```

Functionality Illustration

```
./ipk-l2l3-scan -i eth0 -w 1000 -s 192.168.0.5/25 -s 192.168.0.128/29
-s fd00:cafe:0000:face::1/126
```

Scanning ranges:

192.168.0.0/25 126

192.168.0.128/29 6

fd00:cafe:0000:face::0/126 3

192.168.0.1 arp OK (00-50-56-f1-c7-1b), icmpv4 OK

192.168.0.2 arp OK (00-22-14-ec-46-bb), icmpv4 FAIL

192.168.0.3 arp FAIL, icmpv4 FAIL

...

fd00:cafe:0000:face::1 ndp OK (00-50-56-f1-c7-1b), icmpv6 OK

fd00:cafe:0000:face::2 ndp FAIL, icmpv6 FAIL

fd00:cafe:0000:face::3 ndp OK (a8-5e-45-af-7c-60), icmpv6 FAIL

Output Format

The application is going to be subject to automated testing. It is of utmost importance for the application to write the result to `stdout` exactly as specified.

Program output (`stdout`) consists of 2 ordered sections: 1) scanning ranges summary, then 2) scan results. These sections are separated by an empty line.

The scan ranges summary section starts with a literal `Scanning ranges:` single line. The following lines in this section (individual lines in any order) each contain two values separated by space (): IP address of scanned network (IPv4 or IPv6 with prefix length), and number of usable hosts in the network.

The scan results section consists of one or more lines. Individual lines can be in any order. Each line starts with the scanned host IP address followed by `arp` and ARP/ND scan, then followed by `,` and ICMP/ICMPv6 scan. Scan results must be marked by `arp / ndp / icmpv4 / icmpv6` literals. ARP/ND scan result is either `OK` followed by space with MAC address in parentheses or `FAIL`. ICMP/ICMPv6 scan result is either `OK` or `FAIL`.

```
./ipk-l2l3-scan -i eth0 -s 192.168.0.1/30
```

```
Scanning ranges:
```

```
192.168.0.0/30 2
```

```
192.168.0.1 arp OK (00-50-56-f1-c7-1b), icmpv4 OK
```

```
192.168.0.2 arp OK (00-22-14-ec-46-bb), icmpv4 FAIL
```

```
./ipk-l2l3-scan -i eth0 -s 192.168.0.1/30 -s 192.168.0.130/29
```

```
Scanning ranges:
```

```
192.168.0.0/30 2
```

```
192.168.0.128/29 6
```

```
192.168.0.1 arp OK (00-50-56-f1-c7-1b), icmpv4 OK
```

```
192.168.0.2 arp OK (00-22-14-ec-46-bb), icmpv4 FAIL
```

```
192.168.0.129 arp FAIL, icmpv4 FAIL
```

```
192.168.0.130 arp FAIL, icmpv4 FAIL
```

```
192.168.0.131 arp FAIL, icmpv4 FAIL
```

```
192.168.0.132 arp FAIL, icmpv4 FAIL
```

```
192.168.0.133 arp FAIL, icmpv4 FAIL
```

```
192.168.0.134 arp FAIL, icmpv4 FAIL
```

Bibliography

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